Dr A.C.E. (Arlinde) Vrooman

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Address: Oudezijds Achterburgwal 185, 1012 DK, Amsterdam

E-mail: arlinde.vrooman@huygens.knaw.nl

Profile

My research combines economic history, quantitative and qualitative historical analysis, economics and computational methods, with particular interests in climate, health and disease, colonial West Africa, economics education and the analysis of historical sources. In my recent work, I have increasingly applied Natural Language Processing (NLP), text-as-data, and machine learning methods to large historical and contemporary corpora, alongside more traditional archival and statistical approaches. I also have experience in university teaching and thesis supervision across economics and business, economic history, and interdisciplinary programmes.

Current position

Postdoctoral Researcher, Huygens Institute, KNAW Dec 2025 - present

Postdoctoral Researcher within the project HEAT ("Historische Energietransities en AI-Toepassingen"), at the Department of Data & Collections (Data Management) and affiliated with the LivesLab Research Group.

Previous academic positions

• **Postdoctoral Researcher, Tilburg University** Dec 2023 - Nov 2025

Postdoctoral Researcher at the Tilburg School of Economics and Management (TiSEM): Department of Strategy & Entrepreneurship (as of August 2024), and the Tilburg Center of the Learning Sciences (TiCeLS) and University College Tilburg (UCT) (both as of December 2023).

• **PhD Student, University of Groningen (RUG), Faculty of Economics and Business** Sept 2018 - Mar 2024

Department of Economics, Econometrics and Finance, Economic History Group, and affiliated with the Groningen Growth and Development Center (GGDC) and the Center for Public Health in Economics and Business (CPHEB).

• **Student Assistant, Erasmus University** Oct 2017 - Mar 2018

Assisting research within the Erasmus Centre for Urban, Port and Transport Economics.

Education

PhD in Economics (Economic History), University of Groningen	2018-2024
Posthumus Institute Advanced PhD Training Programme, N.W. Posthumus Institute	2023
Posthumus Institute Basic PhD Training Programme, N.W. Posthumus Institute	2018-2021
MSc in Economics and Business: International Economics, Erasmus University, *Cum Laude*	2017-2018
Honours BSc in Economics and Business Economics, Utrecht University, *Cum Laude*	2013-2016
Selective Joint Excellence Programme in Economics and Business, LUISS University, Rome, *GPA: 4.0/4.0*	2014-2015

Publications

• Vrooman, A.C.E. (2025). The colonial response to the development of disease in Ghana and Côte d'Ivoire (ca. 1900-1955): A comparative analysis of British and French colonial health policies. *PLoS ONE, 20*(8): e0329713. https://doi.org/10.1371/journal.pone.0329713

• Vrooman, A.C.E. (2024). *Here to Heal? The Effect of Colonial Rule on Disease and Health Care in Ghana and Côte d'Ivoire: ca. 1900-1955* (Doctoral dissertation, University of Groningen). https://doi.org/10.33612/diss.902625484

• Vrooman, A.C.E. (2023). The Development of Colonial Health Care Provision in Ghana and Côte d'Ivoire: ca. 1900-1955. *Economic History of Developing Regions, 38*(3), 215-255. https://doi.org/10.1080/20780389.2023.2209284

Books

• Vrooman, A.C.E. *Economic History of Health Care and Disease in Colonial West Africa: The Cases of Ghana and Côte d'Ivoire.* Routledge, Perspectives in Economic and Social History Book Series. Forthcoming in 2027 (Accepted book manuscript).

Working papers

• Vrooman, A. C. E. (2026). Contextual Transformer-Based Disambiguation of Historical Dutch *weder*: An Early Modern Case Study using GLOBALISE. Zenodo. https://doi.org/10.5281/zenodo.23010930
• Vrooman, A.C.E. (2026). Constructing Semantically Coherent and Interpretable Historical Vocabularies from Domain-Specific Corpora: An Early Modern Dutch Case Study in Climate and Weather using VOC archives. Under review at *Digital Humanities Quarterly*. https://doi.org/10.5281/zenodo.20629251
• Vrooman, A.C.E. (2026). Structuring Colonial Health Care Facilities: Understanding the Factors Behind their Spatial Diffusion in Ghana and Côte d'Ivoire, ca. 1900-1950. R&R at *Explorations in Economic History*.
https://dx.doi.org/10.2139/ssrn.6446654
• Vrooman, A.C.E. (2025). Curriculum Development in Academic Economics Education: Evidence from Dutch BSc Programmes in Economics and Business Economics (ca. 2010-2025) using Natural Language Processing. R&R at *International Review of Economics Education*. https://dx.doi.org/10.2139/ssrn.6994920

Research data & other outputs

• Vrooman, A.C.E. (2026). Constructing a Dutch Historical Climate Vocabulary (v1.2) based on GLOBALISE. Zenodo.
https://doi.org/10.5281/zenodo.20134102
• Vrooman, A.C.E. (2026). Constructing a Dutch Historical Climate Vocabulary (v1.1) based on GLOBALISE. Zenodo.
https://doi.org/10.5281/zenodo.19608045
• Vrooman, A.C.E. (2026). Constructing a Dutch Historical Climate Vocabulary (v1) based on GLOBALISE. Zenodo.
https://doi.org/10.5281/zenodo.18246268
• Vrooman, A.C.E. (2024, May 15). The Development of Colonial Health Care Provision in Ghana and Côte d'Ivoire, 1900-1955. *Frontiers in African Economic History*. https://www.aehnetwork.org/blog/the-development-of-colonial-health-care-provision-in-ghana-and-cote-divoire-1900-1955/

Other ongoing research

1. A.C.E. Vrooman. A Natural Language Processing (NLP) approach for analysing curriculum development based on online course guides: topic modelling and n-gram analysis. Submitted to *MethodsX*.
2. A.C.E. Vrooman, J.J.H.M. Brouwer, G. van der Laan. Shaping Future Economists: A Curriculum Mapping Tool for Comparative Analysis of Economics Bachelor Programmes across Dutch Universities.

Research funding & resources

2026: NWO Computing Time on National Computer Facilities. SURF Compute Call 2026. Grant no. EINF-19724. Awarded: 20.000 CPU hrs, 1.000 GPU hrs and 1TB storage (Research Cloud/HPC Cloud).
2020: N.W. Posthumus PhD Grant (€3000).

Teaching experience

2025: Bachelor thesis supervision (840900-B-18), Philosophy of History and Memory (840028-B-6), UCT, BSc Liberal Arts and Sciences.
2024: Bachelor thesis supervision (840900-B-18), Institutional Economics (840806-B-6 & 700403-B-6), Individuals and Collectives: Introduction to Business and Economics and Social Sciences (840114-B-6), UCT, BSc Liberal Arts and Sciences.
2023: Growth, Institutions and Business (EBB022A05), RUG BSc Economics & Business Economics.
2022: Global Business History (EBB038B05), RUG BSc International Business.

Conference, workshop and seminar contributions

2026: Tensions of Europe Conference 2026, TU/e, Eindhoven (presenter); 8th Conference on Digital Humanities and Digital History, C²DH, Esch-sur-Alzette (presenter); Dag van de Nieuwste Geschiedenis, BVGN, Antwerpen (presenter & workshop organizer).
2024: S&E Workshop Research on Education, TiSEM, Tilburg (presenter); TiCeLS Research Seminar, Tilburg (presenter); Annual N.W. Posthumus Conference, Leeuwarden (presenter).
2023: FEB PhD Conference, Groningen (presenter); RHI Seminar, WUR, Wageningen (presenter).
2022: GGDC PhD Workshop, Groningen (presenter); 15th Annual Meeting of the African Economic History Network,

Gran Canaria (presenter); XIX World Economic History Congress, Paris (presenter); European Historical Economics Society Conference, Groningen (presenter).
2021: Annual N.W. Posthumus Conference, online (presenter).
2020: N.W. Posthumus Research Design Course, online (presenter & commentator); Annual N.W. Posthumus Conference, online (commentator).
2019: N.W. Posthumus PhD Seminar 2, Antwerp (presenter & commentator).
2018: N.W. Posthumus PhD Seminar 1, Brussels (presenter & commentator).

Other academic activities

• Peer-reviewing: Economic History of Developing Regions, Medical History, Economics & Human Biology, Journal of Open Humanities Data, Digital Humanities Quarterly.
• 2026: Summer School, Introduction to Historical Linguistics, Leiden University; co-organizer and presenter at workshop in environmental and climate history and public lecture by prof. Dagomar Degroot (Huygens Institute).
• 2019: Co-organizer seminar series in Political Economy, Growth and Development, and Economic History (PEG-DECH), RUG.

Skills

Programming & data: Python, R, Stata, QGIS, ArcGIS, SQL/SQLite, LaTeX, GAMS, Qualtrics.
Python: pandas, NumPy, scikit-learn, SciPy, Gensim, NLTK, Matplotlib, Selenium, fastText, Hugging Face Transformers, PyTorch, regex.
Methods: NLP; text-as-data; information extraction; topic modelling; word embeddings; Transformer-based classification; statistical and econometric analysis; quantitative and qualitative historical analysis; spatial analysis; historical data construction; data extraction from unstructured and structured sources.
Languages: Dutch, English, French, Italian.